

BHUVANA GOPALAM

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EDUCATION

University of Minnesota - Twin Cities

Sep. 2025 – May 2027 (Expected)

Master of Science in Data Science | GPA: 3.68/4

Minneapolis, MN

- **Relevant Coursework:** Data Mining, Database Systems, Applied Regression, Multivariate Analysis, Machine Learning

Mahatma Gandhi Institute of Technology

Dec. 2021 – Jun. 2025

Bachelor of Technology in Data Science | GPA: 8.84/10

Hyderabad, India

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Deep Learning, Big Data Analytics, Natural Language Processing, Predictive Analytics, Database Systems, Operating Systems, Cyber Security, 3D Printing, Computer Networks

EXPERIENCE

Undergraduate Research Assistant - Realtime Violence Detection

Aug. 2024 - May 2025

Mahatma Gandhi Institute of Technology

Hyderabad, India

- Built a real-time violence detection system in PyTorch/OpenCV with **sub-second latency** for large-scale public safety use cases.
- Engineered a high-throughput video pipeline using FFmpeg and GStreamer for scalable dataset ingestion and preprocessing.
- Optimized U-Net + ResNeXt101 + TSM to reduce compute while retaining temporal accuracy for action recognition.
- Deployed an edge-optimized, multi-stream inference stack using TensorRT, InfluxDB, and Streamlit.

PROJECTS

Bone Fracture Detection in Musculoskeletal Radiographs | PyTorch, Object Detection, Medical Imaging | GitHub

- Developed and evaluated **Faster R-CNN** and **RetinaNet** models for fracture localization on the FracAtlas dataset (4,083 radiologist-annotated X-ray images).
- Implemented a **transfer learning pipeline** using ImageNet-pretrained backbones and compared performance against models trained from scratch under identical train/validation/test splits.
- Processed COCO-format bounding box annotations and evaluated detection quality using **IoU, mAP@0.5, precision, and recall** on a held-out test set.
- Achieved **mAP@0.5 = 0.67** and precision = 0.86 using pretrained RetinaNet, outperforming non-pretrained baselines.
- Performed extensive **Grad-CAM interpretability analysis** to visualize model attention regions and assess alignment with clinically annotated fracture sites.
- Identified failure modes including low recall and mislocalized activations, discussing implications for safety-critical clinical deployment.

Self-Questioning Visual Question Answering System | PyTorch, Transformers, OFA, Qwen2-VL, Gradio, llama.cpp

- Designed an explainable cannabis detection system for compliance monitoring using **vision-language** models (OFA, Qwen2-VL) performing reasoning-based inference.
- Built a structured **prompt-driven VQA pipeline** using PyTorch, Transformers, and Fairseq for domain-specific detection and general image analysis.
- Developed an interactive **Gradio web application** deployed via Bash automation and llama.cpp, providing human-readable detection justifications with interpretability outputs.
- Achieved **90% detection accuracy with zero false positives** on the evaluation set, demonstrating robustness for automated compliance screening.

Airline Delay Risk Prediction | Python, Pandas, Random Forest, LightGBM, Feature Engineering

- Built a predictive model using **Random Forest** and **LightGBM** to identify high-delay airline operation periods at the **airport-carrier-month** level, optimizing resource allocation and operational planning.
- Processed and modeled **170K+** aggregated flight records using Pandas and NumPy, handling missing data and class imbalance to improve data quality and model robustness.
- Engineered **15+** features including traffic volume, carrier market share, airport congestion metrics, and seasonality indicators to capture operational delay patterns.

- Evaluated models using cross-validation and confusion matrix analysis, achieving **0.85 ROC-AUC** with LightGBM after correcting data leakage through improved train-test splitting.

Semantic Matching System for Drug Review Data Integration | *PostgreSQL, Python, LLM, Transformers, Pinecone*

- Built a system to link **3100** patient drug reviews with standardized drug data, improving match accuracy from **8%** to **75%** using **semantic similarity**.
- Processed **1500** drugs using embedding based similarity search to handle messy inputs such as brand names, typos, and informal text.
- Designed a **PostgreSQL** database to combine structured drug information with patient reviews, enabling efficient queries on ratings and side effects.
- Enabled analysis such as top side effects per drug and brand versus **generic** comparisons from previously disconnected datasets.

Image Forgery Detection System | *PyTorch, Faster-ViT, EfficientNet-B3, Grad-CAM*

- Built a deepfake detection system using **Faster-ViT** and **EfficientNet-B3** architectures for digital forensics applications.
- Achieved **82.32% validation accuracy** and **81.77% test accuracy** on large-scale datasets, evaluated using confusion matrix analysis, precision (0.64), recall (1.0), and F1-score (0.78).
- Deployed a production-ready **Gradio interface** with Grad-CAM visualization for real-time forgery detection and interpretable attention heatmaps.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, C, Java

Tools & Technologies: Git, Docker, Tableau, Microsoft Excel, AWS Linux, Gradio, VS Code, Anaconda, FastAPI

Frameworks: PyTorch, TensorFlow, Scikit-learn, Transformers, Fairseq, XGBoost, LightGBM, LangChain, LlamaIndex

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Plotly, OpenCV, Pillow (PIL), NLTK, torchvision, llama.cpp

Specializations: Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, MLOps

CERTIFICATIONS

- Google Data Analytics Professional Certificate - Google
- Introduction to Machine Learning, Deep Learning, and Computer Vision - Indian Institute of Technology, Madras
- Data Visualization and Analytics using Tableau - ExcelR

ACHIEVEMENTS & LEADERSHIP

- Awarded Silver Medal for securing 2nd Overall Rank in the Data Science Department during undergraduate studies
- Secured 1st place in Data Science Department during Undergraduate Year 1, ranked by GPA
- Successfully organized and coordinated academic poster presentation during college technical fest
- Coordinated sponsor outreach and funding arrangements for college programs